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Polymerized stroma-free hemoglobin nanoparticle covalently coated with poly(2-ethyl-2-oxazoline) as a red blood cell substitute☆

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ABSTRACT

The development of artificial O_2 carriers as red blood cell (RBC) substitutes has attracted considerable interest, both as a critical resource for emergency medical response and as a complementary tool in routine clinical practice. Building upon the benefits of previously engineered hemoglobin (Hb)-based O_2 carriers, we have designed a core–shell nanoparticle consisting of polymerized stroma-free hemoglobin (SFHb) as the core, enveloped by a covalently attached poly(2-ethyl-2-oxazoline) (POx) shell. This construct, referred to as POx-SFHbNP, exhibits a mean hydrodynamic diameter of approximately 30 nm, with each particle comprising roughly ten Hb molecules in the core and around fifty POx chains in the shell. Notably, the core retains trace amounts of catalase, an enzyme that catalyzes the disproportionation of hydrogen peroxide (H_2O_2), thereby effectively mitigating oxidation of the ferrous heme moieties in Hb. The O_2 complex of POx-SFHbNP remains stable even in aqueous H_2O_2 environments. While the O_2 affinity of POx-SFHbNP exceeds that of native Hb and RBCs, this characteristic may offer therapeutic advantages for O_2 delivery to hypoxic tissues. The formulation exhibited no cytotoxicity and demonstrated excellent hemocompatibility. In a rat model of 50 % hemorrhagic shock, resuscitation with POx-SFHbNP solution achieved 100 % survival at 2 h, accompanied by rapid restoration of blood pressure and heart rate. Serum biochemical analyses indicated no detrimental effects on hepatic or renal function. These results suggest that POx-SFHbNP is a highly promising candidate for use as a RBC substitute.

1. Introduction

Over the past several decades, extensive research has been dedicated to the development of red blood cell (RBC) substitutes [1–5]. Among the various approaches, hemoglobin (Hb)-based O_2 carriers (HBOCs) have remained at the forefront, encompassing strategies such as intramolecularly crosslinked Hbs [6,7], polymerized Hbs [8–11], polyethylene glycol (PEG)-conjugated Hbs (PEG-Hb) [12–15], Hb–albumin clusters [16–18], Hb nanoparticles [19], and Hb-encapsulated vesicles [20,21]. Although some of these formulations have progressed to clinical trials, their widespread application has been impeded by adverse effects, including vasoconstriction, hypertension, and myocardial infarction [22–24]. PEG-Hb has been widely studied and remains under clinical investigation [25,26]. PEG is a hydrophilic polymer recognized for its excellent biocompatibility and has been broadly utilized in various biomedical contexts [27]. PEGylation—covalent attachment of PEG chains to drugs, peptides, and proteins—can enhance their plasma half-life, reduce immunogenicity, and improve pharmacokinetic profiles. However, accumulating evidence suggests that administration of PEGylated therapeutics may elicit the formation of anti-PEG antibodies [28,29]. Subsequent re-administration of PEG-modified agents in individuals with pre-existing antibodies can lead to the accelerated blood clearance (ABC) phenomenon, characterized by rapid systemic elimination, reduced therapeutic efficacy, and potential adverse outcomes [30]. Additionally, repeated dosing with PEGylated proteins has been associated with renal vacuolization [31]. PEG possesses several intrinsic limitations, including synthetic complexity, a requirement for low-temperature storage (−20 °C), the generation of peroxides, elevated viscosity, and a tendency for *in vivo* accumulation—particularly the latter two characteristics being dependent on molecular weight [32,33].

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To address these limitations, we previously synthesized a Hb conjugate with poly(2-ethyl-2-oxazoline) (POx), designated as POx-Hb, and demonstrated its efficacy as a potential artificial O_2 carrier [34,35]. POx has emerged as a promising alternative to PEG owing to its non-immunogenic profile, synthetic fleixibility, structural tunability, and low solution viscosity [32,33,36–39]. Over the past decade, the biocompatibility of POx as a protein-coating material has been extensively validated [32,33,37–39]. Notably, POx exhibits rapid renal clearance without detectable accumulation in organs or tissues [40,41].

Another major challenge in the development of HBOCs is the irreversible oxidation of heme iron from the ferrous (Fe^{2+}) to the ferric (Fe^{3+}) state, which renders Hb incapable of binding O_2 [42]. In contrast to native RBCs, many HBOC formulations lack inherent antioxidant mechanisms, making the suppression of methemoglobin (metHb) formation a critical concern. To addressed this issue, Chang et al. co-polymerized Hb with catalase (Cat)—an enzyme that decomposes hydrogen peroxide (H_2O_2) into H_2O and O_2—and superoxide dismutase (SOD), which catalyzes the dismutation of superoxide radical anion ($O_2^{\bullet-}$). The resulting protein composite effectively inhibited metHb formation [43]. Building upon this concept, we previously developed a core–shell structured nanoparticle by polymerizing stroma-free Hb (SFHb) and covalently attaching human serum albumin (HSA) to the surface, yielding HSA-SFHbNP [44,45]. This nanoparticle, approximately 30 nm in diameter, retains trace levels of Cat within the core, thereby conferring intrinsic antioxidant activity. In the present study, we report the synthesis of a novel POx-conjugated SFHb nanoparticle (POx-SFHbNP), which integrates two synergistic strategies: surface modification with biocompatible POx and core–shell nanoparticle architecture to confer antioxidant functionality. We describe in detail the structural features, O_2 binding properties, and hemocompatibility of the resulting construct. Compared to HSA-SFHbNP, which employs a human albumin coating, POx-SFHbNP offers two distinct advantages: (i) its solution viscosity remains within physiologically acceptable limits even at elevated Hb concentrations, and (ii) it exhibits no species specificity, rendering it potentially applicable not only in humans but also in non-human animals, including companion species such as cats and dogs. Utilizing an anesthetized rat model of hemorrhagic shock, we assessed resuscitative efficacy of POx-SFHbNP by monitoring the recovery of systemic physiological parameters. Comprehensive *in vitro* and *in vivo* evaluations demonstrate that POx-SFHbNP is a safe and effective RBC substitute.

2. Results and discussion

2.1. Synthesis and structural characterization of POx-SFHbNP

SFHb was isolated from the lysate of hemolyzed porcine RBCs by tangential flow filtration (TFF) employing two types of membrane systems (pore-size 0.22 μm and molecular weight cutoff (MWCO) 300 kDa) [45]. The resulting red filtrate, representing SFHb solution, contained trace amounts of Cat. The nanoparticle was engineered to have an average diameter of approximately 30 nm, a size range selected to evade rapid clearance by the reticuloendothelial system [45]. The synthetic strategy is outlined in Fig. 1. Initially, *N*-succinimidyl 3-maleimidopropionate (SMP) was added to the SFHb solution in phosphate-buffered saline (1 × PBS, pH 7.4) under a carbon monoxide (CO) atmosphere to yield maleimide-activated SFHb (MA-SFHb). The MA-SFHb was subsequently polymerized using dithiothreitol (DTT) as a crosslinker, resulting in the formation of MA-SFHb nanoparticle (MA-SFHbNP). Following nanoparticle formation, poly(2-ethyl-2-oxazoline) with a terminal sulfhydryl group (POx-SH, Mw: 5 kDa) [34] was covalently attached to the residual maleimide moieties on the particle surface, affording the final POx-SFHbNP construct. After extensive optimization, the molar ratios of [DTT]/[Hb] = 2.0 and [POx-SH]/[Hb] = 7.0 were identified as the optimal conditions for nanoparticle synthesis. Size exclusion chromatography (SEC) of the reaction mixture revealed a single broad peak eluting at 15–25 min, consistent with the formation of high molecular weight species (Fig. 2A). Unreacted POx-SH was effectively removed via TFF at 4 °C, yielding POx-SFHbNP with a high overall recovery rate (91 %). The use of a 70 kDa MWCO membrane minimized nanoparticle loss and ensured high product purity, with no detectable formation of metHb (metHb ratio: 0 %).

The average hydrodynamic diameter of POx-SFHbNP, as determined by dynamic light scattering (DLS), was 30.7 ± 0.9 nm (Fig. S1A). Vigorous stirring at 300 rpm during the polymerization of SFHb facilitated the formation of monodisperse nanoparticles with a narrow size distribution. Transmission electron microscopy (TEM) analysis of air-dried samples confirmed the presence of uniform, spherical particles with diameters approximating 30 nm (Fig. 2B, Fig. S1B). The nanoparticles exhibited excellent colloidal stability, showing no signs of degradation or aggregation, and their size remained constant over a 6-month storage period. Given that the average diameter of MA-SFHbNP (prior to POx conjugation) was 23.1 nm, the thickness of the hydrated POx shell was estimated to be approximately 3.8 nm, consistent with previously reported values for POx-Hb [34,35]. Nanoparticles of this size readily pass through standard 0.2 μm sterilizing-grade filters, enabling efficient sterile filtration—a critical advantage in pharmaceutical manufacturing and formulation. The circular dichroism (CD) spectrum of POx-SFHbNP was identical to those of unmodified Hb and SFHb at equivalent concentrations (Fig. 2C), indicating that neither the polymerization process nor POx conjugation perturbed the secondary structure of the Hb molecule.

Static light scattering (SLS) measurements estimated the weight-average molecular weight of MA-SFHbNP to be approximately 664 kDa, corresponding to an average of 10.3 Hb molecules per nanoparticle. Quantitative analysis of Hb content combined with gravimetric evaluation of lyophilized samples yielded a [POx]/[Hb] molar ratio of 4.8. These findings suggest that each POx-SFHbNP consists of a core

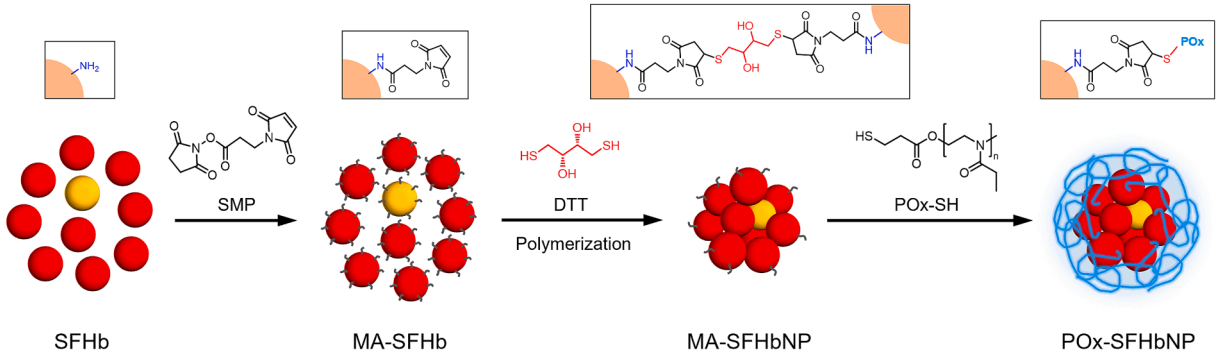


Fig. 1. Schematic illustration of the synthesis of POx-SFHbNP. Red spheres represent Hb, and yellow spheres indicate Cat. Abbreviations are defined in the main text. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

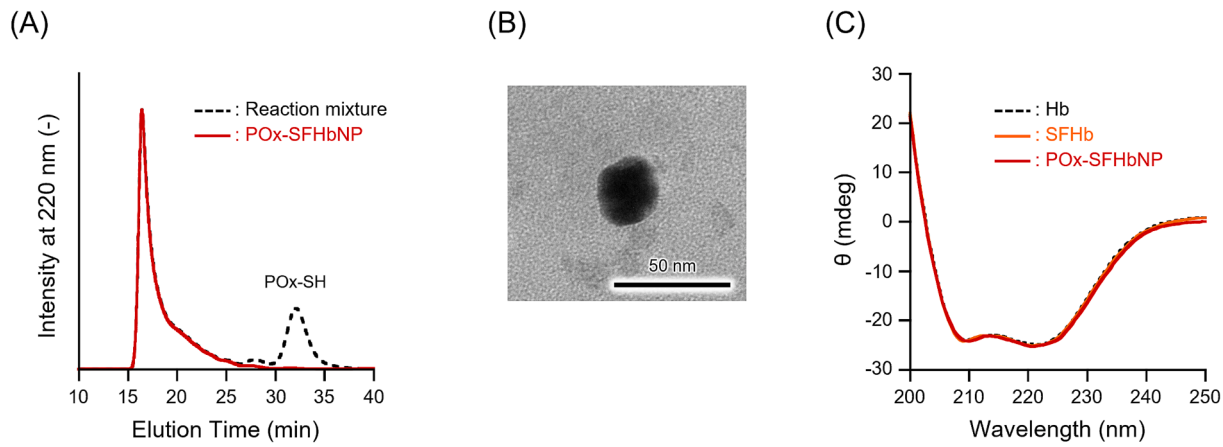


Fig. 2. (A) SEC profile of POx-SFHbNP at 25 °C. (B) TEM image of POx-SFHbNP. (C) CD spectra of POx-SFHbNP and SFHb (in PBS, pH 7.4, 25 °C).

containing roughly 10 Hb molecules, enveloped within a shell formed by approximately 50 POx chains, resulting in an estimated total molecular weight of 915 kDa.

Given that the isoelectric point of Hb is approximately 7.0, the zeta potential of MA-SFHbNP measured at pH 7.0 was 0.05 mV. Surface coating of these nanoparticles with the nonionic, water-soluble polymer POx effectively masked the surface-exposed amino acid residues; however, this modification did not substantially alter the apparent zeta potential, which was measured at −0.11 mV for POx-SFHbNP. The colloid osmotic pressure (COP) of POx-SFHbNP solution exhibited a concentration-dependent increase, reaching 13.3 mmHg at an Hb concentration of 5 g dL^{-1} (Fig. S1C). This value exceeds that of the MA-SFHbNP solution at the same Hb concentration, yet remains slightly lower than the COP of a 5 g dL^{-1} HSA solution (18.1 mmHg), indicating that POx-SFHbNP possesses a suitably moderate COP, favorable for physiological application.

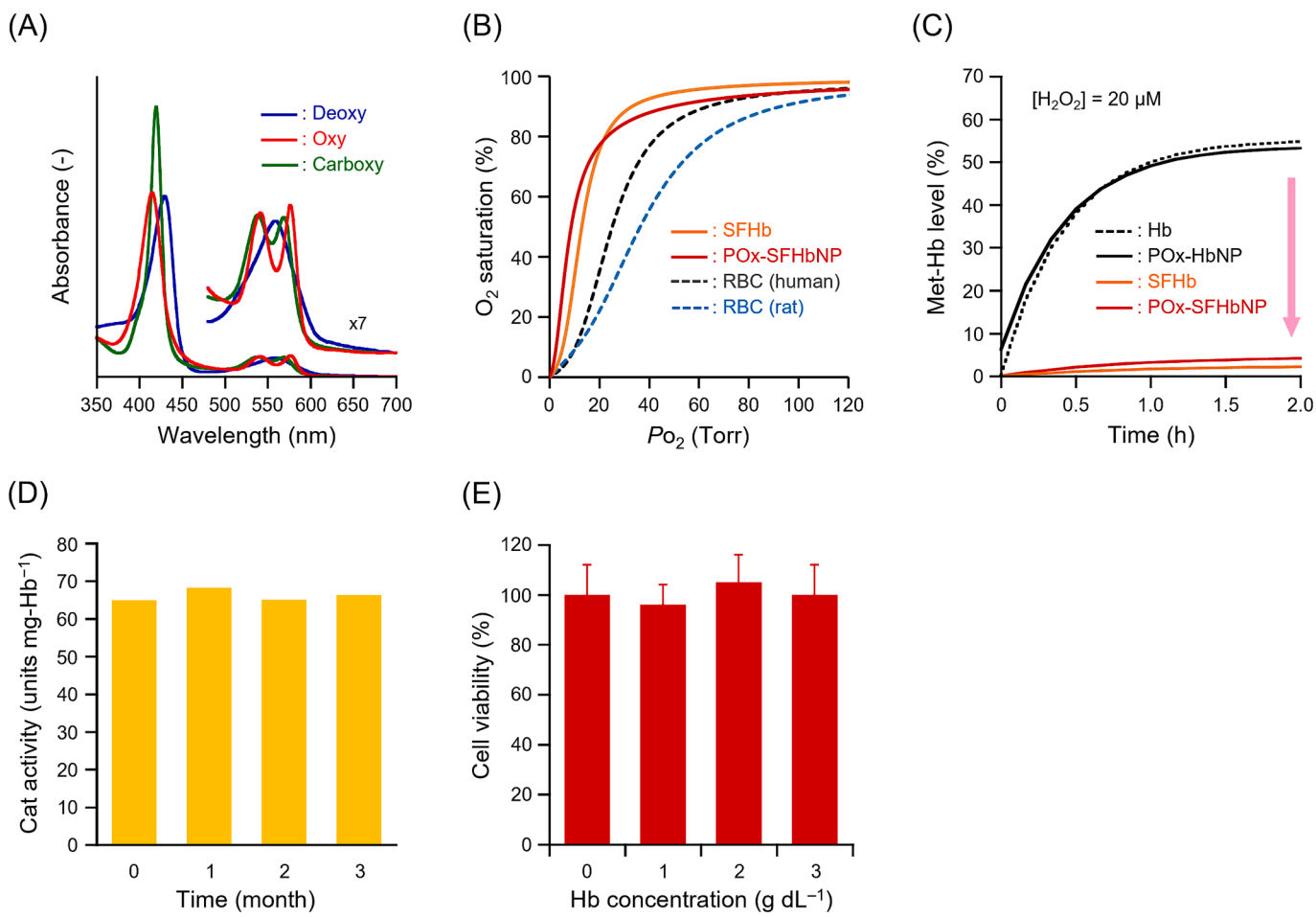


Fig. 3. (A) Visible absorption spectral changes of POx-SFHbNP (in PBS, pH 7.4, 25 °C). The magnified (×7) Q bands were shown in the inset. (B) O_2 dissociation curves of POx-SFHbNP, SFHb, RBCs (human and rat) (in PBS, pH 7.4, 37 °C). (C) Time-dependent changes in metHb levels in POx-SFHbNP and POx-HbNP in 20 μM H_2O_2 solution ([Hb] = 10 μM) at 25 °C. (D) Stability of Cat activity in POx-SFHbNP during storage at 4 °C. (E) Correlation between Hb concentration in POx-SFHbNP and cell viability of HUVECs following exposure to POx-SFHbNP. Each value presents the mean ± SD (n = 3).

The shear rate–viscosity profile of the POx-SFHbNP solution ([Hb] = 5 g dL^{-1}) exhibited Newtonian fluid behavior, with a viscosity of 2.2 cP at a shear rate of 230 s^{-1} (Fig. S2). This value is lower than that of human whole blood (Hct 45 %, 3.7 cP) and the HSA-SFHbNP solution at the same Hb concentration (3.6 cP). Notably, even at an increased Hb concentration of 10 g dL^{-1}, the viscosity of the POx-SFHbNP solution remained at 5.6 cP, which falls within the physiologically acceptable range. In contrast, the viscosity of the HSA-SFHbNP at [Hb] = 10 g dL^{-1} rose markedly to 27.3 cP. The comparatively low viscosity of the POx-SFHbNP solution is attributable to the low intrinsic viscosity of the conjugated POx, a favorable characteristic for clinical application.

2.2. O_2 binding property

The visible absorption spectra of POx-SFHbNP solution (in PBS, pH 7.4) under N_2, O_2, and CO atmospheres—corresponding to the deoxy, oxy, and carboxy forms of Hb, respectively—closely resembled those of unmodified Hb in the same states (Fig. 3A). These results indicate that neither nanoparticle formation nor covalent polymer modification altered the electronic structure of the heme moiety.

The O_2 dissociation curve of POx-SFHbNP (Fig. 3B) yielded a P_{50} value (the partial pressure of O_2 at which 50 % of Hb is saturated) of 9 Torr and a Hill coefficient (*n*, indicative of cooperative binding) of 1.4 at 37 °C (in PBS, pH 7.4) (Table 1). In comparison, the parent SFHb exhibited a P_{50} of 13 Torr and a Hill coefficient of 2.5, suggesting that POx-SFHbNP possesses a slightly higher O_2 affinity and reduced cooperativity. We have previously demonstrated that modification of Hb with SMP, which reacts with Cys-β93 and surface lysine residues, inhibits the quaternary transition from the relaxed (R) to the tense (T) state, thereby increasing O_2 affinity [16,17,34,35,44,45]. The enhanced O_2 affinity and diminished cooperativity observed in POx-SFHbNP are likely attributable to the same mechanism. These findings are consistent with those reported by Vandegriff [12], Manjula [13], Li [14], and Styslinger [46]. The MA-SFHbNP precursor, prior to POx coating, exhibited P_{50} and *n* values nearly identical to those of the final construct, indicating that the subsequent attachment of POx chains did not affect the O_2 affinity of the SFHb nanoparticle (Table 1). This observation is consistent with previously reported data for HSA-SFHbNP [45]. Importantly, an elevated O_2 affinity is not inherently disadvantageous. Under hypoxic conditions, such as those encountered in hemorrhagic shock, higher O_2 affinity may actually enhance tissue oxygenation by promoting efficient O_2 uptake and retention during systemic circulation.

2.3. Stability of the O_2 complex

The stability of the O_2 complex in oxy-POx-SFHbNP (in PBS, pH 7.4) was assessed by measuring the autoxidation rate constant (k_{ox}) of the heme iron within the Hb component at 37 °C. For comparison, POx-HbNP—prepared using purified Hb and lacking endogenous enzymes—was synthesized under analogous conditions. The k_{ox} value for POx-HbNP was 0.056 h^{-1}, whereas that for POx-SFHbNP was markedly lower at 0.038 h^{-1}, indicating significant suppression of Hb oxidation.

Table 1
O_2 binding parameters of POx-SFHbNP, related Hb derivatives, and RBCs (in PBS, pH 7.4, 37 °C).

	P_{50} (Torr)	*n* (–)
SFHb (human)[a]	12	2.4
SFHb (porcine)	13	2.5
MA-SFHbNP	8	1.4
POx-SFHbNP	9	1.4
HSA-SFHbNP[a]	8	1.4
RBC (human)[a]	25	2.5
RBC (porcine)	33	2.7
RBC (rat)	35	2.3

[a] From ref. [45].

This enhanced stability is likely attributable to residual Cat retained within the polymerized SFHb core, which serves as an antioxidant for oxy-Hb. The uncoated MA-SFHbNP exhibited a k_{ox} of 0.033 h^{-1}, suggesting that the POx conjugation does not significantly influence the redox behavior of the SFHb core.

During Hb autoxidation, $O_2^{\bullet-}$ and H_2O_2 are generated, both of which further accelerate oxidation of the heme iron [42]. Due to the short half-life of $O_2^{\bullet-}$, the removal of H_2O_2 is considered particularly effective in mitigating autoxidation in HBOCs [43,45,47]. Physiological concentrations of H_2O_2 in healthy human blood range from 1 to 36 μM [48]. Based on previous findings with HSA-SFHbNP, we hypothesized that Cat retained in the SFHb core of POx-SFHbNP would similarly inhibit oxyHb oxidation. To evaluate this, the stability of the O_2 complex was tested in an aqueous H_2O_2 solution (20 μM) at 25 °C. Time-dependent absorbance changes at 630 nm—corresponding to metHb—revealed a pronounced difference between POx-HbNP and POx-SFHbNP (Fig. 3C). In both POx-HbNP and unmodified pure Hb, 53–55 % of oxy-Hb was oxidized within 2 h, likely due to the preferential oxidation of the α chains, which are more susceptible than β chains [49]. In contrast, POx-SFHbNP and SFHb displayed exceptional stability, showing negligible oxidation in the same conditions. After 2 h, the metHb levels remained as low as 2–4 %. These results demonstrate that POx-SFHbNP effectively retain the intrinsic antioxidant capacity of SFHb. Consequently, POx-SFHbNP may exert antioxidant activity *in vivo*, suggesting a dual role as both O_2 carrier and antioxidant agent.

2.4. Cat activity

To further elucidate the mechanism underlying the antioxidant capacity of POx-SFHbNP, we evaluated the enzymatic activity of Cat retained within the SFHb core by measuring the rate of H_2O_2 decomposition. In our previous work, we observed that SOD in SFHb is either removed or inactivated during purification and polymerization processes [44,45].

Upon the addition of POx-SFHbNP to an aqueous H_2O_2 solution in phosphate buffer (PB, pH 7.0), the characteristic absorbance of H_2O_2 at 240 nm decreased over time. The Cat activity of POx-SFHbNP was determined to be 65 ± 1 units mg-Hb^{-1}, which is slightly lower than that of the starting SFHb (85 ± 4 units mg-Hb^{-1}), yet remains comparable. These results confirm that the antioxidant function of POx-SFHbNP primarily derives from the potent H_2O_2-degrading activity of the co-retained Cat. The enzymatic activity is preserved even after the polymerization process. A likely explanation for the retention of Cat activity is that the catalytic heme sites are embedded within its large globular structure, rendering them less susceptible to chemical modification. In future studies, we also intend to explore the use of specialized Hb solutions supplemented with defined amounts of purified Cat to enhance antioxidant functionality. The Cat activity of POx-SFHbNP remained stable after storage in PBS at 4 °C for 3 months at an Hb concentration of 5 g dL^{-1} (Fig. 3D). With this confirmed, we proceeded to evaluate the safety and efficacy of POx-SFHbNP as a RBC substitutes through a series of *in vitro* and *in vivo* experiments.

2.5. Cytotoxicity and hemocompatibility

To evaluate the cytotoxicity of POx-SFHbNP, we investigated its effect on human umbilical vein endothelial cells (HUVECs). Cells were incubated with oxy-POx-SFHbNP at various Hb concentrations ([Hb] = 1, 2, and 3 %) for 6 h. Cell viability remained unaffected at all concentrations tested, indicating that POx-SFHbNP exhibits no significant cytotoxicity toward vascular endothelial cells (Fig. 3E).

Hemocompatibility was further assessed by mixing POx-SFHbNP ([Hb] = 5 g dL^{-1}) with freshly collected whole blood *in vitro* at varying volume ratios (9/1, 8/2, and 6/4, v/v). After mixing, the counts of RBCs, white blood cells (WBCs), and platelets (PLTs) were quantified. Depending on the dilution ratio, the cell counts decreased to

approximately 90 %, 80 %, and 60 % of the initial values, respectively, and remained stable over a 4-h incubation period at 37 °C (Fig. S3). These results demonstrate that POx-SFHbNP is non-cytotoxic and exhibit favorable compatibility with blood cellular components. Such properties support its potential as a safe and viable RBC substitute for *in vivo* applications.

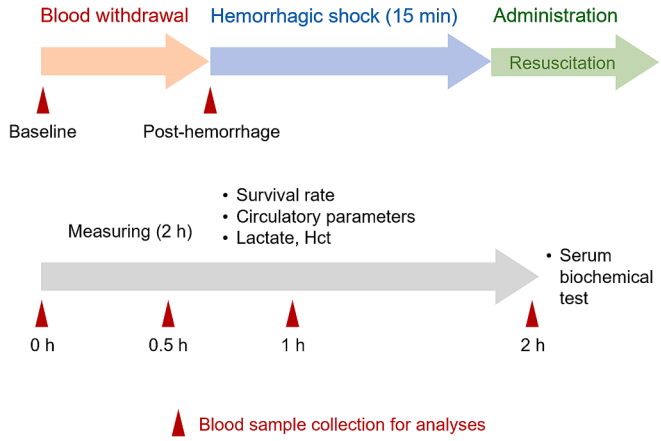
2.6. Resuscitation from hemorrhagic shock

To evaluate the efficacy of POx-SFHbNP solution as an RBC substitute, we employed a rat model of hemorrhagic shock using 50 vol% exchange transfusion, a well-established method for assessing resuscitative performance (Scheme 1) [18,34,45]. Hemorrhagic shock was induced by withdrawing 50 % of the total blood volume. After a 15-min stabilization period, resuscitation was carried out by intravenous administration of POx-SFHbNP solution ([Hb] = 5 g dL^{-1}, in PBS, pH 7.4) in a volume equal to that of the withdrawn blood. To provide a comparative evaluation, control groups included rats resuscitated with shed autologous blood (SAB) and the Ringer's lactate solution (RL). Survival rates and circulatory parameters were monitored for 2 h following injection. Baseline physiological values prior to blood withdrawal served as reference. At the conclusion of the experiment, blood samples were collected for serum biochemical analysis and animals were euthanized. This experimental protocol enables a comprehensive assessment of the safety and efficacy of POx-SFHbNP in restoring systemic circulation and organ function under hemorrhagic shock.

2.6.1. Survival rate, circulatory parameters, lactate levels, and hematocrit values

As anticipated, all animals in the POx-SFHbNP and SAB groups survived for the full 2-h observation period following administration (Fig. 4A). In contrast, 60 % of animals in the RL group died within 2 h, indicating that the RL solution alone failed to provide adequate O_2 delivery for recovery from severe hemorrhagic shock.

Before blood withdrawal, the mean arterial pressure (MAP) was 112 ± 7 mmHg (Fig. 4B). Following the 50 vol% blood loss, MAP dropped sharply to 28 ± 2 mmHg. In the POx-SFHbNP group, MAP rapidly returned to baseline level (120 ± 9 mmHg) after administration and remained stable at approximately 95 mmHg after 2 h. The SAB group showed a similar immediate recovery (130 ± 4 mmHg) although MAP gradually declined thereafter. In contrast, the RL group exhibited persistently low MAP, never recovering to baseline and reaching only 44 mmHg at 2 h. It is well-established that free Hb scavenges nitric oxide (NO), an endothelium-derived vasodilator, potentially resulting in acute hypertension [22,50]. Notably, no hypertensive response was observed in the POx-SFHbNP group, suggesting that the nanoparticles remained intravascular and did not freely diffuse into the vessel wall to interact with NO.



Scheme 1. Experimental protocol for hemorrhagic shock and resuscitation in rats. Timeline depicts the 2-h observation period following resuscitation with POx-SFHbNP after 50% blood loss induction.

Heart rate (HR) also declined from 409 ± 20 to 331 ± 19 beats min^{-1} following blood loss, indicating bradycardia (Fig. 4C). POx-SFHbNP administration restored HR to baseline within 1 h. Conversely, the RL group exhibited poor HR recovery throughout the 1.5-h observation window. These results indicate that POx-SFHbNP is effective in reversing bradycardia and stabilizing circulatory function.

Hemorrhagic shock induced anaerobic metabolism, reflected in a lactate increase from 1.0 ± 0.3 to 5.4 ± 0.7 mmol L^{-1} (Fig. 4D). In both the POx-SFHbNP and SAB groups, lactate levels returned to baselines within 30 min post-administration, suggesting effective O_2 delivery to peripheral tissues. In contrast, lactate level in the RL group remained elevated (high variability is due to animals that succumbed during the observation period).

The hematocrit (Hct) value dropped to 30 ± 1 % immediately after blood withdrawal (Fig. 4E). Administration of SAB, which contains RBCs, restored Hct value to baseline within 30 min. On the other hand, the acellular POx-SFHbNP solution further reduced Hct value to 17 ± 1 %, with no significant change over 2 h. Despite this, the particles remained intravascular and contributed to increased circulating volume, acting as an effective plasma expander with O_2 carrying capability. In the RL group, Hct value initially fell to 23 ± 2 % and then rose slightly to 27 % by 2 h. This partial recovery may reflect hemoconcentration due to lack of COP in the RL solution, leading to plasma extravasation and reduced blood volume. Overall, these results underscore the ability of POx-SFHbNP to serve dually as an O_2 carrier and plasma expander, providing superior resuscitative performance compared to RL solution in hemorrhagic shock.

2.6.2. Serum biochemical analyses

Serum biochemical analyses were conducted using arterial blood samples collected 2 h after administration. For the RL group, results were based on the two surviving animals. Baseline (control) values were obtained from blood samples taken before hemorrhagic shock induction.

Aspartate aminotransferase (AST) and alanine aminotransferase (ALT) serve as biomarkers of hepatic injury, as their levels increase in response to liver cell damage. AST levels in the POx-SFHbNP group (73 ± 19 units L^{-1}) and SAB group (58 ± 7 units L^{-1}) were comparable to the baseline (Fig. 5A), indicating preserved liver function in both groups. The ALT level in the POx-SFHbNP group (55 ± 7 units L^{-1}) was slightly elevated compared to the baseline (33 ± 6 units L^{-1}) but did not differ significantly from the SAB group (58 ± 9 units L^{-1}) (Fig. 5B). Lactate dehydrogenase (LDH), a broader marker of tissue injury including liver, heart, kidney, and blood, remained within normal range in the POx-SFHbNP group (124 ± 43 units L^{-1}), similar to the baseline (91 ± 27 units L^{-1}) and the SAB group (108 ± 17 units L^{-1}) (Fig. 5C). In stark contrast, the RL group exhibited significantly elevated levels of AST (232 ± 70 units L^{-1}), ALT (199 ± 77 units L^{-1}), and LDH (1538 ± 736 units L^{-1}), suggestive of substantial hepatic and systemic tissue damage, likely associated with ischemia–reperfusion injury following inadequate O_2 resuscitation. These findings indicate that POx-SFHbNP did not induce hepatic dysfunction, as evidenced by the maintenance of serum enzyme levels within physiological ranges.

Blood urea nitrogen (BUN) and uric acid (UA) are important indicators of renal function. In the POx-SFHbNP group, the BUN level (23 ± 5 mg dL^{-1}) was comparable to the baseline and significantly lower than that observed in the SAB group (31 ± 3 mg dL^{-1}) (Fig. 5D). Moreover, the UA levels in both the POx-SFHbNP group and the SAB group were below baseline levels (Fig. 5E), suggesting no induction of renal stress or dysfunction. In contrast, the RL group showed markedly elevated UA level relative to baseline, indicating potential kidney injury, likely due to reperfusion stress following inadequate O_2 delivery. These results imply that POx-SFHbNP does not place strain on the kidneys. We concluded that the POx-SFHbNP solution did not induce any detrimental

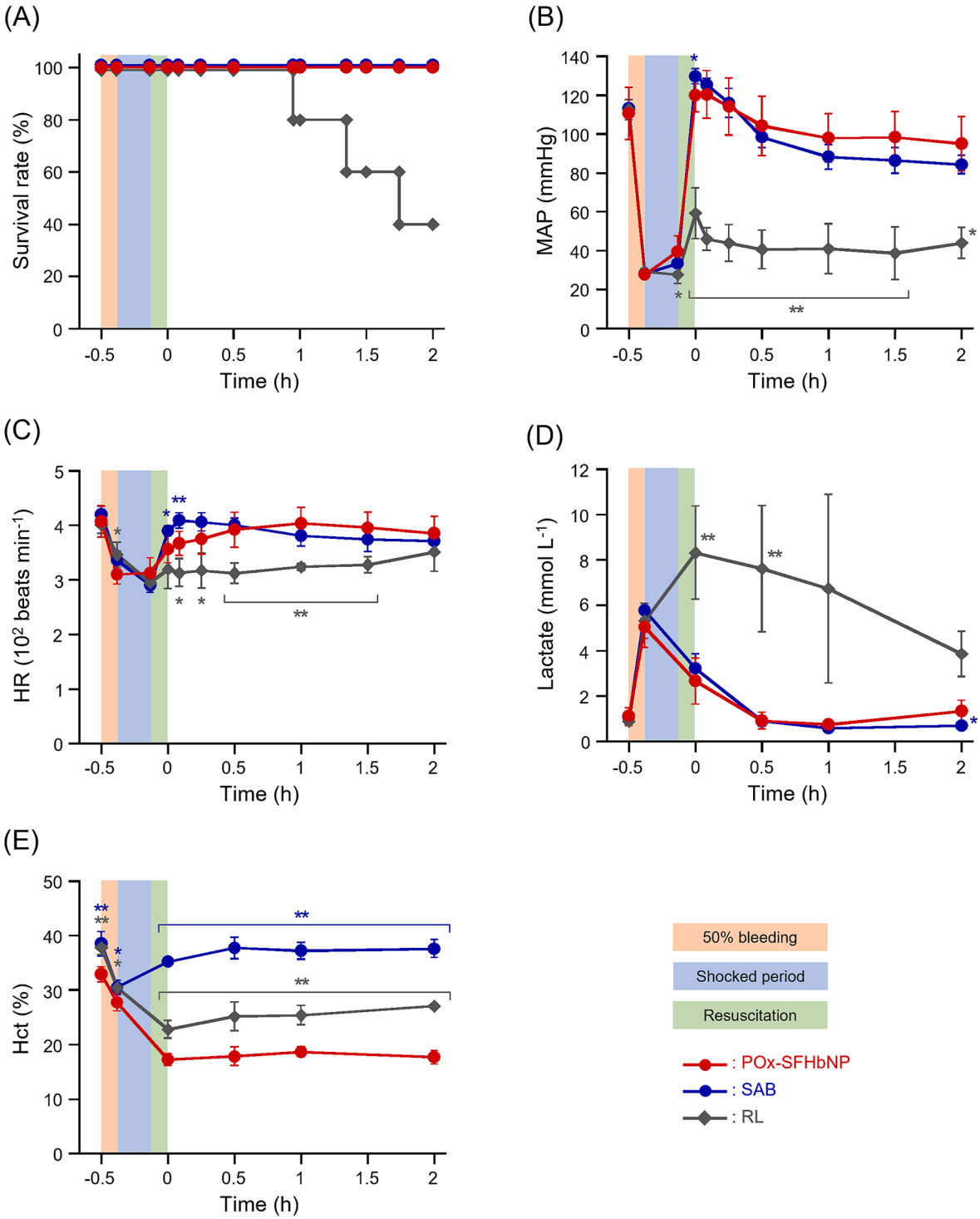


Fig. 4. Changes in (A) survival rate, (B) MAP, (C) HR, (D) blood lactate level, and (E) Hct in anesthetized rats during hemorrhagic shock and subsequent resuscitation with POx-SFHbNP solution ([Hb] = 5 g dL^{-1}, in PBS, pH 7.4). Each data point represents the mean ± SD (n = 6). Control data for the SAB and RL groups (n = 5 each) were obtained from previous experiments conducted under identical conditions [34]. *P < 0.05, **P < 0.01 versus POx-SFHbNP group. Data for the RL group after 1, 1.5, and 2 h reflect only surviving animals.

effects on hepatic or renal function, underscoring its potential safety in critical conditions such as hemorrhagic shock. The resuscitative efficacy of the POx-SFHbNP solution was comparable to that of both the HSA-SFHbNP and POx-Hb formulations in the same rat models of 50 % hemorrhagic shock [34,45].

3. Conclusions

POx-SFHbNP was successfully synthesized by covalently attaching POx chains to the surface of polymerized SFHb particles with high yield. DLS measurements and TEM analyses confirmed the formation of uniform nanoparticles with an average diameter of approximately 30 nm. The resulting POx-SFHbNP exhibited moderate O_2 affinity (P_{50} = 9 Torr) and antioxidant capacity attributed to the retained Cat enzyme within

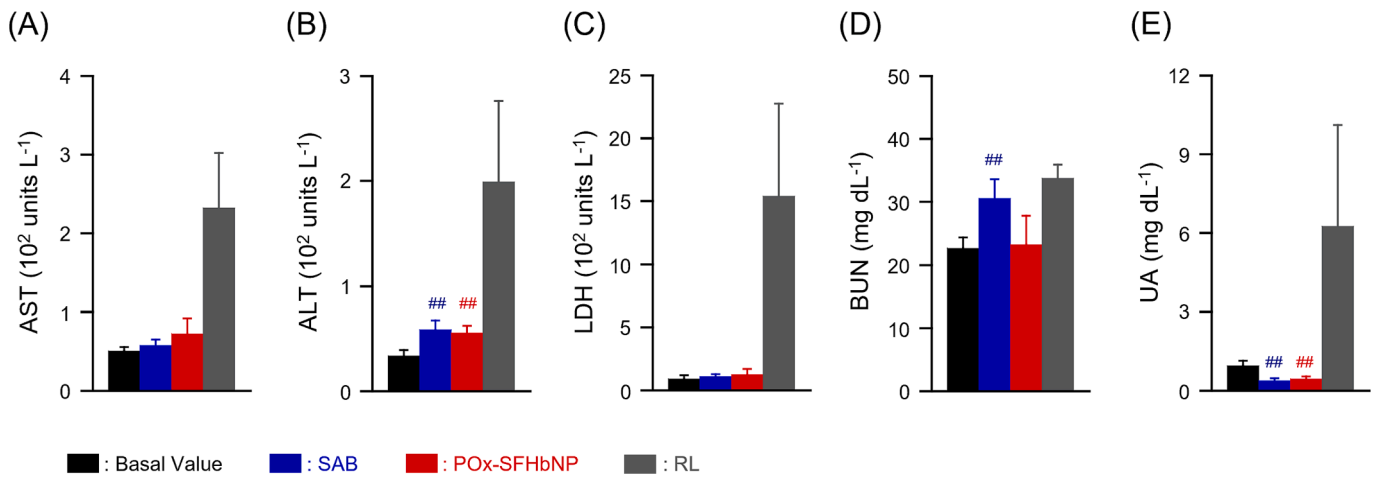


Fig. 5. Serum biochemical parameters at 2 h after resuscitation with POx-SFHbNP solution ([Hb] = 5 g dL^{-1}, in PBS, pH 7.4) in anesthetized rats: (A) AST, (B) ALT, (C) LDH, (D) BUN, and (E) UA. Each value represents the mean ± SD (n = 6). Control data for the basal, SAB, and RL groups (n = 5 each) were obtained from previous experiments performed under identical conditions [34]. ##P < 0.01 versus basal value. Data for the RL group reflect two surviving rats at 2 h time point.

the SFHb core. The POx-SFHbNP solution showed no cytotoxic effects on the endothelial cells and demonstrated good compatibility with blood cell components. In a rat model of 50 % hemorrhagic shock, intravenous administration of POx-SFHbNP led to rapid restoration of blood pressure, heart rate, and lactate levels—comparable to recovery achieved by transfusion of shed autologous blood. Furthermore, serum biochemical analyses revealed no adverse effects on liver or kidney function. These findings demonstrate that POx-SFHbNP is not only an effective O_2 carrier but also acts as a plasma expander with antioxidant properties. The formulation shows considerable promise as a safe and effective RBC substitute for resuscitation in hemorrhagic shock and potentially broader clinical applications.

4. Experimental section

4.1. Material and apparatus

Fresh porcine blood from specific pathogen-free (SPF) swine was obtained from Tokyo Shibaura Zouki Co. Ltd. Porcine SFHb was prepared from porcine RBCs using a modified version of the previously reported method for human SFHb [45]. The procedure for purifying porcine Hb from RBCs was followed as described in earlier literature [34]. Hydroxyl-terminated poly(2-ethyl-2-oxazoline) (Mw; 5 kDa, PDI ≤ 1.3) was purchased from Merck KGaA. Sulfhydryl-terminated poly(2-ethyl-2-oxazoline) (POx-SH) was synthesized following our previously reported protocol [34,51]. *N*-Succinimidyl 3-maleimidopropionate (SMP) was obtained from Tokyo Chemical Industry Co., Ltd., and dithiothreitol (DTT) from Fujifilm Wako Pure Chemical Corp. All other reagents were of analytical grade, purchased from commercial suppliers, and used without further purification. Deionized water (18.2 MΩ cm) was prepared using a water purification system (Milli-Q IQ7003; Merck KGaA). Unless otherwise specified, the concentration of phosphate-buffered saline (PBS) used was 1×, consisting of 137 mM NaCl, 2.7 mM KCl, 10 mM Na_2HPO_4, and 1.76 mM KH_2PO_4, adjusted to pH 7.4. Ringer's lactate (RL) solution (Lactec Injection) was supplied by Otsuka Pharmaceutical Co., Ltd. UV–Vis absorption spectra were recorded using a UV–Visible spectrophotometer (V-750; JASCO Corp.) at 25 °C. CD spectra were obtained using a CD spectrophotometer (J-1100; Jasco Corp.) at 25 °C. Additional details on TEM, COP, and viscosity measurements are provided in the Supplementary material.

4.2. Synthesis of POx-SFHbNP

A DMSO solution of SMP (200 mM, 1.5 mL) was added dropwise to a PBS solution of carboxy SFHb (1 mM, 15 mL) in a 50-mL round-bottom flask at a rate of 1 mL min^{-1} ([SMP]/[Hb] = 20 mol/mol). The mixture was stirred for 30 min at 25 °C under CO atmosphere in the dark. Following the reaction, the mixture was centrifuged at 14,000 g for 5 min at 4 °C. The supernatant was subjected to GFC (Sephadex G-25 superfine; Cytiva) using PBS as the elution buffer to remove excess SMP. The eluate containing MA-SFHb was adjusted to 45 mL ([Hb] = 333 μM) with PBS. A freshly prepared PBS solution of DTT (8.3 mM, 3.61 mL, [DTT]/[Hb] = 2.0 mol/mol) was then added dropwise to the MA-SFHb solution at a rate of 2 mL min^{-1} while stirring vigorously at 300 rpm. The reaction was continued for 60 min at 25 °C in the dark. Next, a PBS solution of POx-SH (4.0 mM, 26.4 mL, [POx-SH]/[Hb] = 7.0 mol/mol) was added dropwise at 6 mL min^{-1}, and the resulting dispersion was stirred at 300 rpm for 18 h at 4 °C under CO atmosphere in the dark. The final reaction mixture (total volume: 75 mL, [Hb] = 0.2 mM) was subjected to TFF (Omega PES Membrane T-series TFF cassettes 70 kDa; Pall Corp.) at 4 °C to remove unreacted POx-SH by dialysis against deionized water. The resulting aqueous POx-SFHbNP solution was then concentrated using TFF to ≥ 816 μM and changed to 1 × PBS solution (pH 7.4, [Hb] = 5 g dL^{-1}) by adding 5.3 % (v/v) of 20 × PBS. The successful formation of POx-SFHbNP was confirmed via SEC using an HPLC system equipped with a Superdex 200 Increase column (Cytiva), with PBS as the mobile phase. The final preparation was sequentially sterilized using syringe filter units (DISMIC 45CS, 0.45 μm pore, followed by DISMIC 25CS, 0.2 μm pore; Tokyo Roshi Kaisha Ltd.), transferred into glass vials, sealed, filled with CO gas, and stored in the dark at 4 °C.

For comparison, enzyme-free POx-HbNP with a diameter of approximately 30 nm was also synthesized from purified porcine Hb using the same procedure.

4.3. Particle size, zeta potential, molecular weight, and [POx]/[Hb] ratio

The hydrodynamic diameter and zeta potential of POx-SFHbNP were determined by dynamic light scattering (DLS) using a zeta-potential and particle analyzer (ELSZ-2000ZS; Otsuka Electronics Co., Ltd.). The weight-average molecular weight of MA-SFHbNP was evaluated by static light scattering (SLS) measurements performed with the same instrument. Measurements were conducted in PBS (pH 7.4) at sample concentrations ranging from 1.0 to 4.0 mg mL^{-1}. The molecular weight was calculated using a Debye plot, assuming a differential refractive index increment (dn/dc) of 0.185.

The molar ratio of POx to Hb ([POx]/[Hb]) in POx-SFHbNP was determined by gravimetric analysis of the freeze-dried sample. An aqueous POx-SFHbNP solution ([Hb] = 0.8 mM, 0.25 mL, in deionized

water) was lyophilized for 18 h under reduced pressure using a freeze dryer (FDU-1200; Tokyo Rikakikai Co. Ltd.). The [POx]/[Hb] molar ratio was calculated by subtracting the known weight of Hb (quantified by the cyan-metHb method) from the total weight of the lyophilized powder.

4.4. O_2 binding property, Cat activity, and cytotoxicity

Visible absorption spectra of the deoxy (under N_2), oxy (under O_2), and carboxy (under CO) forms of POx-SFHbNP ([Hb] = 3 μM, in PBS, pH 7.4) were recorded using a UV–visible spectrophotometer (V-750; Jasco Corp.). The O_2 affinity (P_{50} value) and Hill coefficient (n) were determined with an automatic recording system for the O_2-equilibrium curve (Hemox-Analyzer; TCS Scientific Corp.) at 37 °C. The methodology for evaluating O_2 complex stability has been described previously [34,44]. Cat activity within POx-SFHbNP was assessed according to previously reported protocols [44].

The cytotoxicity of POx-SFHbNP was assessed using HUVEC cells (immortalized human umbilical vein endothelium cell line, HUEhT-1; JCRB Cell Bank) following a previously established procedure [35]. All experiments were conducted in triplicate for each group (n = 3). Data are presented as mean ± standard deviation (SD).

4.5. Animal experiments and recovery from hemorrhagic shock

Male Wistar rats were obtained from Charles River Laboratories Japan, Inc. and acclimated for one week under a controlled 12-h light/dark cycle with free access to food and water. All animal procedures were conducted in accordance with ARRIVE guidelines [52] and approved by the Animal Care and Use Committee of Saitama Medical University (3394). Animal care complied with the National Institutes of Health (NIH) guidelines for the care and use of laboratory animals. Hemocompatibility testing protocols are provided in the Supplementary material.

Hemorrhagic shock experiments were performed using male Wistar rats (8 weeks old, 241 ± 8 g, n = 6), following previously reported procedures [34]. Additional methodological details are available in the Supplementary material. Data are expressed as mean ± standard deviation (SD). Statistical significance between groups was determined using two-tailed unpaired Student's *t*-test; $p < 0.05$ was considered significant.

CRediT authorship contribution statement

Natsumi Kohyama: Writing – original draft, Methodology, Investigation, Validation. **Tomoka Araki:** Writing – original draft, Investigation. **Mayuka Fujita:** Investigation. **Junya Fujisawa:** Investigation. **Taiga Yamada:** Investigation. **Masatoshi Yamaguchi:** Investigation. **Mitsutomo Kohno:** Writing – review & editing, Investigation. **Teruyuki Komatsu:** Writing – original draft, Investigation, Validation, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.eurpolymj.2025.114126.

Data availability

The data supporting the findings of this study are available in the Supplementary material and from the corresponding author upon reasonable request.

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